CNGB3-Related Retinopathy (Achromatopsia Type 3): Comprehensive Disease Report

Target disease: CNGB3-Related Retinopathy — MONDO:0100446 — OMIM #262300 (Achromatopsia 3) **Causal gene: CNGB3** (cyclic nucleotide-gated channel beta 3; HGNC:2153; NCBI Gene 54714; Ensembl ENSG00000170289; UniProt Q9NQW8; chromosome 8q21.3), encoding the β (modulatory) subunit of the cone cyclic nucleotide-gated (CNG) cation channel.

Summary

CNGB3-related retinopathy is a congenital, autosomal recessive cone dysfunction syndrome ("achromatopsia," ACHM) caused by biallelic loss-of-function variants in *CNGB3*, the gene encoding the modulatory β-subunit of the cone cyclic nucleotide-gated cation channel. Loss of a functional cone CNG channel abolishes the final step of cone phototransduction, producing the clinical tetrad of severely reduced visual acuity (~20/200), absent or severely impaired color discrimination, photophobia/photoaversion, and infantile (pendular) nystagmus, with a characteristic electroretinogram (ERG) pattern of absent photopic (cone) responses and preserved scotopic (rod) responses. The disease is present from birth and was historically regarded as "stationary," but longitudinal structural and functional imaging demonstrates slow progressive cone loss over years.

CNGB3 is the single most common achromatopsia gene, accounting for approximately 45–50% of all cases, and achromatopsia overall affects roughly 1 in 30,000 people, making it the most common cone dysfunction syndrome. A single frameshift founder allele, **c.1148delC (p.Thr383fs)**, accounts for roughly 70% of all *CNGB3* mutant alleles, which simplifies molecular diagnosis and has important implications for carrier screening and genotype-targeted therapy. The core pathophysiology is a **loss-of-function cone channelopathy**: loss of the channel abolishes the light-sensitive conductance, and disturbed intracellular Ca^{2+}/cGMP–PKG signaling together with endoplasmic reticulum (ER) stress and unfolded-protein-response–

associated apoptosis drives slow cone degeneration while sparing rods. This has been established in *Cngb3*$^{-/-}$ mice, cone-dominant (*Nrl*$^{-/-}$) ER-stress models, and a naturally occurring canine model.

Management is currently **supportive** — tinted/red-filter contact lenses or spectacles to reduce photophobia, refractive correction, low-vision aids, and genetic counseling. The leading experimental therapy is **AAV-mediated *CNGB3* gene replacement**, which rescues cone function in dog and mouse models. The first-in-human trial (AAV8-hCARp.hCNGB3, NCT03001310) demonstrated acceptable safety but **no consistent efficacy signal at 6 months**, highlighting the central translational challenge: the therapeutic window, cone target availability, and functional plasticity of a lifelong-deprived visual system.

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1. Disease Information

Overview. CNGB3-related retinopathy is the *CNGB3* genetic subtype of **achromatopsia (ACHM)**, also called rod monochromatism or complete/incomplete achromatopsia. It is a congenital, autosomal recessive, non-progressive-to-slowly-progressive retinal disorder of isolated cone photoreceptor dysfunction. "Achromatopsia is the most common cone dysfunction syndrome, affecting 1 in 30,000 people. It is an autosomal recessive disorder" (PMID: 39273686).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0100446
OMIM	#262300 (Achromatopsia 3, ACHM3)
Gene	*CNGB3* — HGNC:2153; NCBI Gene 54714; Ensembl ENSG00000170289; UniProt Q9NQW8
Gene locus	8q21.3
Orphanet	ORPHA:49382 (Achromatopsia)
MeSH	Color Vision Defects / Achromatopsia
ICD-10	H53.51 (color vision deficiency / achromatopsia)
ICD-11	9D45 (color vision deficiencies)

Synonyms / alternative names: CNGB3 retinopathy; achromatopsia type 3 (ACHM3); rod monochromacy / rod monochromatism; complete achromatopsia (typical) and incomplete achromatopsia (atypical); total color blindness; cone dysfunction syndrome (CNGB3 subtype).

Information source. The information in this report is derived almost entirely from **aggregated disease-level resources** (OMIM, Orphanet, HPO) and from **human clinical case series/cohort studies** plus **model-organism and in-vitro mechanistic studies** — not from individual EHR records.

2. Etiology

Primary cause — genetic. The disease is monogenic and recessive: it requires two pathogenic alleles (homozygous or compound heterozygous) in *CNGB3*. "CNGB3 encoding the beta subunit of the cyclic nucleotide-gated ion channel in cone photoreceptors is the major achromatopsia gene" (PMID: 28795510). There are **no established environmental, infectious, toxic, or lifestyle causes**; the disease is not acquired.

Genetic risk factors. - The sole causal factor is biallelic pathogenic variation in *CNGB3*. - The predominant allele, **c.1148delC (p.Thr383fs)**, "accounts for over 70% of all CNGB3 mutant alleles" (PMID: 15657609). An equivalent designation, p.T383fsX, "accounted for 75% (18/24) of

disease-associated alleles; intragenic SNPs in unrelated patients revealed transmission of a common haplotype consistent with a founder effect" (PMID: 17265047). - **Consanguinity** increases the chance of homozygosity for recessive alleles; several cohorts comprise consanguineous families (e.g., PMID: 42099125).

Environmental risk factors. None known. Age and sex are not risk factors; the disease is congenital and affects males and females equally.

Protective factors. No genetic or environmental protective factors are established. In principle, inheriting only one pathogenic allele (heterozygous carrier) is non-disease-causing.

Gene–environment interactions. None demonstrated. Bright-light exposure exacerbates the *symptom* of photophobia but is not a disease modifier. There is no evidence that any exposure modifies the rate of cone degeneration.

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3. Phenotypes

The clinical phenotype is highly stereotyped across cohorts. In a genetically confirmed ACHM cohort (n=21), "All patients experienced difficulties with color vision, 20 of 21 (95.2%) had photosensitivity, and 18 of 21 patients reported congenital nystagmus" (PMID: 41867372). Hallmark features (vision loss, photophobia, color blindness, refractive error, nystagmus) are consistently reported across cohorts (PMID: 42099125).

Phenotype	Type	Onset	Severity	Course	Frequency	HPO term
Color vision loss (achromatopsia)	Symptom / lab (color testing)	Congenital	Severe (complete) to moderate (incomplete)	Stable/ slowly progressive	~100%	HP:0011520 (Defective cone cell function); HP:0000551 (Abnormal color vision)
Photophobia / photoaversion	Symptom	Infancy	Moderate–severe	Stable	~95%	HP:0000613 (Photophobia)
Infantile nystagmus (pendular)	Clinical sign	Neonatal/ infancy (first months)	Variable; may dampen with age	Often improves	~85–95%	HP:0001339 (Congenital nystagmus); HP:0000639 (Nystagmus)
Reduced visual acuity (~20/200)	Symptom / lab	Congenital	Severe	Slowly progressive	~100%	HP:0000572 (Visual loss); HP:0007843 (Reduced visual acuity)
Absent/reduced photopic ERG	Lab abnormality	Congenital	Severe	Stable	~100%	HP:0000512 (Abnormal electroretinogram
Refractive error (hyperopia; myopia in some)	Sign	Childhood	Mild–moderate	Stable	Common	HP:0000540 (Hyperopia); HP:0000545 (Myopia)
Foveal hypoplasia / EZ disruption	Structural (OCT)	Congenital	Variable	Slowly progressive	Common	HP:0007750 (Foveal hypoplasia)
Central scotoma / eccentric fixation	Sign	Childhood	Moderate	Stable	Variable	HP:0000603 (Central scotoma)

ERG signature: "most patients with a normal scotopic response and absent photopic response on electroretinogram" (PMID: 41867372) — preserved rod function, absent cone function. This rod/cone dissociation is diagnostic.

Complete vs. incomplete ACHM. Complete (typical) achromatopsia shows total loss of color discrimination and acuity ~20/200; incomplete forms retain some residual color vision and slightly better acuity. *CNGB3* most commonly causes complete achromatopsia.

Quality-of-life impact. Lifelong low vision, disabling photophobia (bright light markedly worsens function), and nystagmus affect education, mobility, inability to drive, and glare in daily life. In the first-in-human gene therapy trial, 21/23 participants reported some patient-reported QoL benefit, underscoring the measurable daily-functioning burden (PMID: 37172884). No disease-specific validated EQ-5D/SF-36 datasets were identified in this investigation.

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4. Genetic / Molecular Information

Causal gene. *CNGB3* (OMIM *605080), 8q21.3, encodes the β (modulatory) subunit of the cone CNG channel. The functional cone channel is a heterotetramer of three CNGA3 (α) + one CNGB3 (β)** subunits. CNGA3 forms the pore; CNGB3 modulates ligand sensitivity, ion permeation, and regulation by calmodulin and phosphoinositides (P 36430626; P 23530136).

Contribution to disease. "Of these, 485 (45.2%) carried mutations in CNGB3" out of 1074 families clinically diagnosed with achromatopsia (PMID: 28795510); independently, "mutations in the CNGB3 gene are responsible for approximately 50% of all patients with achromatopsia" (PMID: 15657609). Combined, "Mutations in the cone CNG channel subunits CNGA3 and CNGB3 account for >70% of all known cases of achromatopsia" (PMID: 22493484).

Pathogenic variants. - **Predominant allele:** c.1148delC (p.Thr383fsX, a.k.a. p.Thr383IlefsX13) — a 1-bp frameshift deletion producing a premature stop and loss of function; >70% of *CNGB3* mutant alleles (PMID: 15657609; PMID: 17265047). - **Variant classes:** predominantly loss-of-function — frameshift, nonsense, splice-site, and large copy-number variations (CNVs/deletions were part of the mutation spectrum reported in PMID: 28795510). Missense variants occur but are a minority. - **Classification:** c.1148delC is **Pathogenic** (ACMG/AMP: null variant in a gene

where LOF is a known mechanism, PVS1 + founder/segregation evidence). - **Allele frequency:** c.1148delC is present at low frequency in gnomAD consistent with a recessive-disease carrier allele; exact population frequencies were not re-derived in this investigation. - **Origin: germline** only; no somatic contribution. - **Functional consequence: loss of function** of the cone CNG channel β-subunit. (By contrast, some *CNGA3* missense variants, e.g., R410W, confer a gain of spontaneous channel activity — see Mechanism.)

Modifier genes. No robust disease-severity modifier genes are established for *CNGB3* ACHM. The phenotype (complete vs. incomplete) is primarily allele-driven.

Epigenetic information. No disease-specific DNA-methylation or histone-modification changes are established for CNGB3 retinopathy.

Chromosomal abnormalities. Large *CNGB3* deletions/CNVs are part of the allelic spectrum (PMID: 28795510), but gross aneuploidy or translocations are not features of this disorder.

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5. Environmental Information

Environmental factors: None causally implicated. This is a monogenic disease. **Lifestyle factors:** None. Bright-light environments worsen photophobic *symptoms* but do not cause or modify the disease. **Infectious agents:** Not applicable — no infectious etiology.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *CNGB3*** (most often c.1148delC) **lead to** absent or non-functional CNGB3 β-subunit protein in cone photoreceptors.
2. Loss of the β-subunit **results in** a non-functional or absent cone CNG channel heterotetramer (3×CNGA3 + 1×CNGB3), impairing channel assembly, trafficking, and gating.
3. The defective channel **abolishes** the light-regulated inward Na^+/Ca^{2+} conductance that normally closes in response to light-driven cGMP hydrolysis — i.e., **loss of the final step of cone phototransduction**.

4. Loss of cone phototransduction **produces** the immediate clinical phenotype: absent photopic ERG, severe color-vision loss, reduced acuity, photophobia, and (via abnormal foveal visual input during development) infantile nystagmus and foveal hypoplasia. *(Rods are spared because they use a distinct CNGA1/CNGB1 channel — scotopic ERG is preserved.)*
5. In parallel, channel deficiency **causes** disturbed intracellular **Ca^{2+}/cGMP homeostasis**. (Branch A) Elevated cGMP/PKG signaling with **RYR2 (ryanodine receptor 2) upregulation leads to** ER Ca^{2+} dysregulation and protein mislocalization. (Branch B) In certain gain-of-function channel contexts, **spontaneously active channels** admit excess Ca^{2+}.
6. ER Ca^{2+} dysregulation and accumulation of mislocalized/misfolded cone proteins **trigger** the **unfolded protein response / ER stress** (↑Grp78/BiP, ↑phospho-eIF2α, ↑CHOP).
7. Sustained ER stress **activates** calpains and caspase-12/caspase-7, **resulting in ER-stress–associated cone apoptosis**.
8. Progressive cone apoptosis **produces** the slow structural cone degeneration (EZ disruption, cone-density decline) observed longitudinally, narrowing the therapeutic window — while rods remain intact.

> Steps 1–4 are well demonstrated in human patients and mouse models. Steps 5–7 are demonstrated in cone-dominant mouse models (*Cng*$^{-/-}$;*Nrl*$^{-/-}$) and in-vitro systems; their quantitative contribution in human CNGB3 retina is inferred.

Supporting detail

Molecular pathways / biochemistry. The affected pathway is **cone phototransduction** (KEGG hsa04744; Reactome "Visual phototransduction"). The cone CNG channel (CNGA3/CNGB3) is the terminal effector. Downstream disturbances involve **cGMP–PKG signaling** and **ER-stress/UPR** (PERK–eIF2α–ATF4–CHOP; IRE1; ATF6). In CNG-deficient cones, "up-regulation of the cysteine protease calpains and cleavage of caspase-12 and caspase-7 were found in the channel-deficient retinas, suggesting an endoplasmic reticulum stress-associated apoptosis" (PMID: 22493484).

Ca^{2+}/cGMP–PKG and RYR2. RYR2 upregulation contributes to cGMP/PKG-signaling-induced cone degeneration (PMID: 32173907) and to impaired protein localization under ER Ca^{2+} dysregulation (PMID: 31182474).

Aberrant channel activity (related channelopathy insight). An achromatopsia *CNGA3* mutation illustrates a complementary gain-of-function route to cone death: "CNGA3_R410W/CNGB3 and TAX4_R421W channels are spontaneously active without cGMP and induce cell

death, suggesting cone degeneration triggered by spontaneous CNG channel activity as a possible cause of achromatopsia" (PMID: 35233102). For *CNGB3* LOF, loss of conductance (not spontaneous activity) is the dominant mechanism.

ER retrotranslocation as a modifiable node. Overexpression of the E3 ubiquitin ligase **synoviolin-1 (SYVN1/HRD1)** promotes ER retrotranslocation, reduces ER stress, and preserves cones in CNG-channel deficiency (PMID: 39215566) — nominating the UPR/ERAD axis as a neuroprotective target.

Cell types & GO/CL terms. - Primary cell type: **cone photoreceptor cell** (CL:0000573); subtypes L/M/S cones. - Biological processes: **detection of light stimulus involved in visual perception / phototransduction** (GO:0007602, GO:0050908); **cGMP-gated cation channel activity** (GO:0005223); **response to ER stress / UPR** (GO:0034976, GO:0030968); **apoptotic process** (GO:0006915). - Chemical entities (CHEBI): cGMP (CHEBI:16356), Ca^{2+} (CHEBI:29108), Na^{+} (CHEBI:29101).

Molecular profiling. In *CNGB3*$^{-/-}$ mice, M-opsin, cone transducin, and cone arrestin were reduced 50–60% by 1 month and cone opsin mislocalized to the ONL/OPL (PMID: 21273547) — a proteomic/trafficking signature of cone stress. No human transcriptomic/proteomic/metabolomic disease signatures were identified in this investigation.

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7. Anatomical Structures Affected

Organ level. Primary organ: the **eye (retina)**; specifically the **neural retina** and the **fovea/macula** (cone-rich central retina). Secondary involvement: oculomotor system (nystagmus). Body system: **visual/nervous system** only. No systemic/extraocular involvement.

Tissue & cell level. Affected tissue: **photoreceptor layer of the neural retina** (nervous tissue). Targeted cells: **cone photoreceptors** (CL:0000573) — L, M, and S cones. **Rods are spared** (scotopic ERG preserved; PMID: 21273547, PMID: 41867372).

Subcellular level (GO cellular component). Cone **outer segment / connecting cilium** plasma membrane (channel location; GO:0001750 photoreceptor outer segment); **endoplasmic reticulum** (GO:0005783) — site of UPR/ER stress; **plasma membrane** (GO:0005886). Opsin mislocalizes to the inner segment/outer nuclear layer in disease.

Localization (UBERON). Retina (UBERON:0000966); fovea centralis (UBERON:0001782); macula lutea (UBERON:0002776); photoreceptor layer / outer nuclear layer (UBERON:0001789). **Lateralization: bilateral and symmetric.**

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8. Temporal Development

Onset. Congenital — cone dysfunction is present from birth. Symptoms (nystagmus, photophobia) are usually noticed within the **first months of life**; onset of initial symptom at a mean of ~0.88 ± 1.07 years in one preschool cohort ( ). P 40241905

Onset pattern is **chronic/insidious** rather than acute.

Progression. Historically classified as "stationary," but modern longitudinal data show **slow functional and structural progression**: - Structurally relatively stable over months: in 41 CNGB3 patients imaged over 6–26 months, "Foveal cone structure showed little or no change in this group of subjects with CNGB3-associated achromatopsia" (EZ grade unchanged in 34/41) (PMID: 28145975). - Functionally slowly progressive over years: "ACHM is a progressive disease in terms of BCVA, MS and ERG responses, and affects slowly the structural integrity of the retina" — BCVA declined ~0.012 logMAR/year and macular sensitivity −0.16 dB/year (PMID: 33562422). - CNGA3 foveal cone mosaic declines ~2%/year, "a similar rate of change to CNGB3-associated ACHM (2% decline per year)" (PMID: 39365261).

Disease course. Lifelong/chronic; non-remitting. There is **no spontaneous remission**.

Critical period / therapeutic window. Because cone structure is relatively preserved early and declines slowly, there is a **finite window** for cone-directed gene therapy, best defined by direct imaging of photoreceptor structure rather than age alone (PMID: 28145975). This argues for early intervention in childhood.

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9. Inheritance and Population

Epidemiology. Achromatopsia overall: **~1 in 30,000** ("Achromatopsia is the most common cone dysfunction syndrome, affecting 1 in 30,000 people"; PMID: 39273686). CNGB3 accounts for ~45–50% of these (PMID: 28795510; PMID: 15657609), implying a CNGB3-specific prevalence on the order of ~1 in 60,000.

Inheritance. Autosomal recessive (AD/X-linked/mitochondrial excluded). "It is an autosomal recessive disorder" (PMID: 39273686).

Penetrance / expressivity. Essentially **complete penetrance** in biallelic individuals; expressivity ranges from complete to incomplete achromatopsia, largely allele-dependent.

Genetic anticipation / mosaicism. Not applicable (no repeat expansion); germline mosaicism not a recognized feature.

Founder effect. Strong — the c.1148delC / p.T383fs allele reflects a **common founder haplotype**: "intragenic SNPs in unrelated patients revealed transmission of a common haplotype consistent with a founder effect" (PMID: 17265047).

Consanguinity. Elevates recessive-homozygote risk; documented in consanguineous cohorts (PMID: 42099125).

Carrier frequency. Consistent with ~1/30,000 ACHM and CNGB3's ~50% share, the population carrier frequency for CNGB3 pathogenic alleles is roughly on the order of ~1/120–1/170 (derived from Hardy-Weinberg; exact gnomAD figure not re-derived here).

Population demographics. Occurs worldwide. The c.1148delC founder allele is especially common in populations of European descent. Specific founder mutations underlie high local prevalence in some isolated/consanguineous populations. **Sex ratio ~1:1** (autosomal). Age distribution: present from birth, all ages affected.

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10. Diagnostics

Electrophysiology — the key functional test. Full-field ERG shows **absent/severely reduced photopic (cone) and 30-Hz flicker responses with preserved scotopic (rod) responses** — the diagnostic signature (PMID: 41867372). Specialized cycle-by-cycle ERG methods can validate very low-amplitude cone responses in advanced disease (P 38558093).

Imaging. Spectral-domain OCT (SD-OCT) grades foveal structure on a 5-point scale (continuous ellipsoid zone → EZ disruption → EZ absence → hyporeflective zone → outer retinal atrophy) and detects foveal hypoplasia (P 24148654).

Adaptive optics scanning light ophthalmoscopy (AOSLO) directly images the cone mosaic / peak cone density (P 39365261 , P 28145975).

Fundus autofluorescence (FAF) and infrared imaging reveal foveal/parafoveal abnormalities. Handheld SD-OCT enables assessment in preverbal children (P 24676353).

Color vision & psychophysics. Color vision testing (e.g., Berson plates, anomaloscope) confirms severe/complete color deficiency; photoaversion testing documents photophobia.

Genetic testing — definitive. Molecular confirmation is essential for diagnosis, prognosis, and gene-therapy eligibility. Recommended approach: - **Targeted ACHM gene panel / NGS panel** covering the six ACHM genes (**CNGA3, CNGB3, GNAT2, PDE6C, PDE6H, ATF6**) is first-line; panels detect SNVs and CNVs. - **Single-variant / single-gene testing** is efficient given the common c.1148delC founder allele (test for it first in appropriate populations). - **Whole-exome sequencing (WES)** is useful for panel-negative cases; periodic reanalysis increases yield (P 39189993).

- **Chromosomal microarray / MLPA** for suspected large CNGB3 deletions. - Electrophysiology-guided genetic testing maximizes molecular diagnosis in pediatric IRD (P 35456422).

Differential diagnosis. Blue cone monochromatism (X-linked, *OPN1LW/OPN1MW*; better acuity, myopia, preserved S-cones — P 40736816); other ACHM subtypes (CNGA3, GNAT2, PDE6C, ATF6 — ATF6 shows near-absent foveal cones, P 31237654; PDE6C often with myopia and progressive maculopathy, P 31826238); cone/cone-rod dystrophy; Leber congenital amaurosis; oculocutaneous albinism and other causes of infantile nystagmus / foveal hypoplasia (P 35157951, P 40232708).

Screening. Carrier screening (especially for the c.1148delC founder allele), cascade testing of relatives, and prenatal/preimplantation genetic testing are available for known-mutation families. Achromatopsia is not part of standard newborn metabolic screening but is often identified during work-up of infantile nystagmus.

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11. Outcome / Prognosis

Survival / mortality. Achromatopsia is **not life-limiting**; life expectancy is **normal**. No disease-specific mortality.

Morbidity / function. Lifelong **visual disability**: best-corrected acuity typically ~20/200, disabling photophobia, inability to drive, dependence on low-vision aids, and impacts on education and daily functioning. Visual disability is stable to slowly worsening.

Disease course / complications. No systemic complications. The main "complication" is slow progressive cone loss (BCVA ~0.012 logMAR/year; PMID: 33562422) which gradually erodes the cone substrate available for future therapy.

Recovery potential. Without treatment, the deficit is permanent. Gene therapy aims to restore/preserve cone function but has not yet shown consistent clinical efficacy (see Treatment).

Prognostic factors. Residual cone structure (EZ integrity, peak cone density on OCT/AOSLO) is a better predictor of therapeutic potential than age (PMID: 28145975). Earlier intervention, before cone loss, is expected to yield better outcomes. Genotype alone does not reliably predict photoreceptor preservation ( ).

P 24676353

12. Treatment

Current standard: supportive care

- **Tinted / red or magenta filter contact lenses or spectacles** to reduce photophobia and improve contrast (NCIT: tinted lens / optical device).
- **Refractive correction** and **low-vision aids** (magnifiers, high-contrast materials).
- **Dark / cutoff sunglasses** outdoors.
- **Genetic counseling** for families (recessive inheritance, carrier testing).

Advanced therapeutics — gene therapy (experimental, lead approach)

Rationale. Because CNGB3 ACHM is a monogenic loss-of-function disease with a slow-degenerating, accessible target (cones), **AAV-mediated CNGB3 gene replacement** is the primary therapeutic strategy. "Up to 80% of the patients carry mutations in the genes CNGA3 and CNGB3... a number of promising gene supplementation therapy programs have been initiated" ( ).

P 28095637

First-in-human trial (NCT03001310). Subretinal **AAV8-hCARp.hCNGB3** in 23 CNGB3-ACHM patients (11 adults, 12 children): "AAV8-hCARp.hCNGB3 (11 adults, 12 children) was safe and generally well tolerated. Intraocular inflammation occurred in 9 of 23 participants" (mostly mild/moderate, two serious dose-limiting events); but "There was no consistent pattern of change from baseline to week 24 for any efficacy assessment" (PMID: 37172884). Individual improvements were seen in color vision (6/23), photoaversion (11/20), and QoL (21/23).

Preclinical support. - **Canine model:** AAV5 expressing human or canine CNGB3 under a cone-opsin promoter "rescued cone photoreceptor function, but at high doses was associated with an inflammatory response (focal chorioretinitis) consistent with immune-mediated toxicity" (PMID: 29020838). - **Mouse model:** *Cngb3*$^{-/-}$ mice recapitulate early-onset, slowly progressive cone dysfunction, providing a platform for therapy (PMID: 21273547).

Neuroprotective / adjunct strategies (preclinical)

- **ER-stress modulation:** SYVN1/synoviolin-1 overexpression reduces ER stress and preserves cones (PMID: 39215566) — a candidate adjunct to prolong the treatment window.
- **Targeting cGMP/PKG–RYR2 signaling** is a proposed route to slow degeneration (P 32173907, P 31182474).

Not applicable

No approved pharmacotherapy, no pharmacogenomics, no surgical / immunotherapy / cell-therapy options are established. No disease-specific drug exists.

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13. Prevention

- **Primary prevention:** Not possible (congenital genetic disease). Prevention is reproductive/genetic: **genetic counseling**, **carrier screening** (particularly for the c.1148delC founder allele), **cascade testing**, and reproductive options including **preimplantation genetic testing (PGT)** and **prenatal diagnosis** for known-mutation families.
- **Secondary prevention:** Early diagnosis via infantile-nystagmus work-up (ERG + genetic testing) enables early low-vision support and candidacy for emerging gene therapy.
- **Tertiary prevention:** Photoprotection (tinted lenses) to maximize residual function and comfort; low-vision rehabilitation; educational accommodations.
- **Immunization / public health / environmental interventions:** Not applicable.

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14. Other Species / Natural Disease

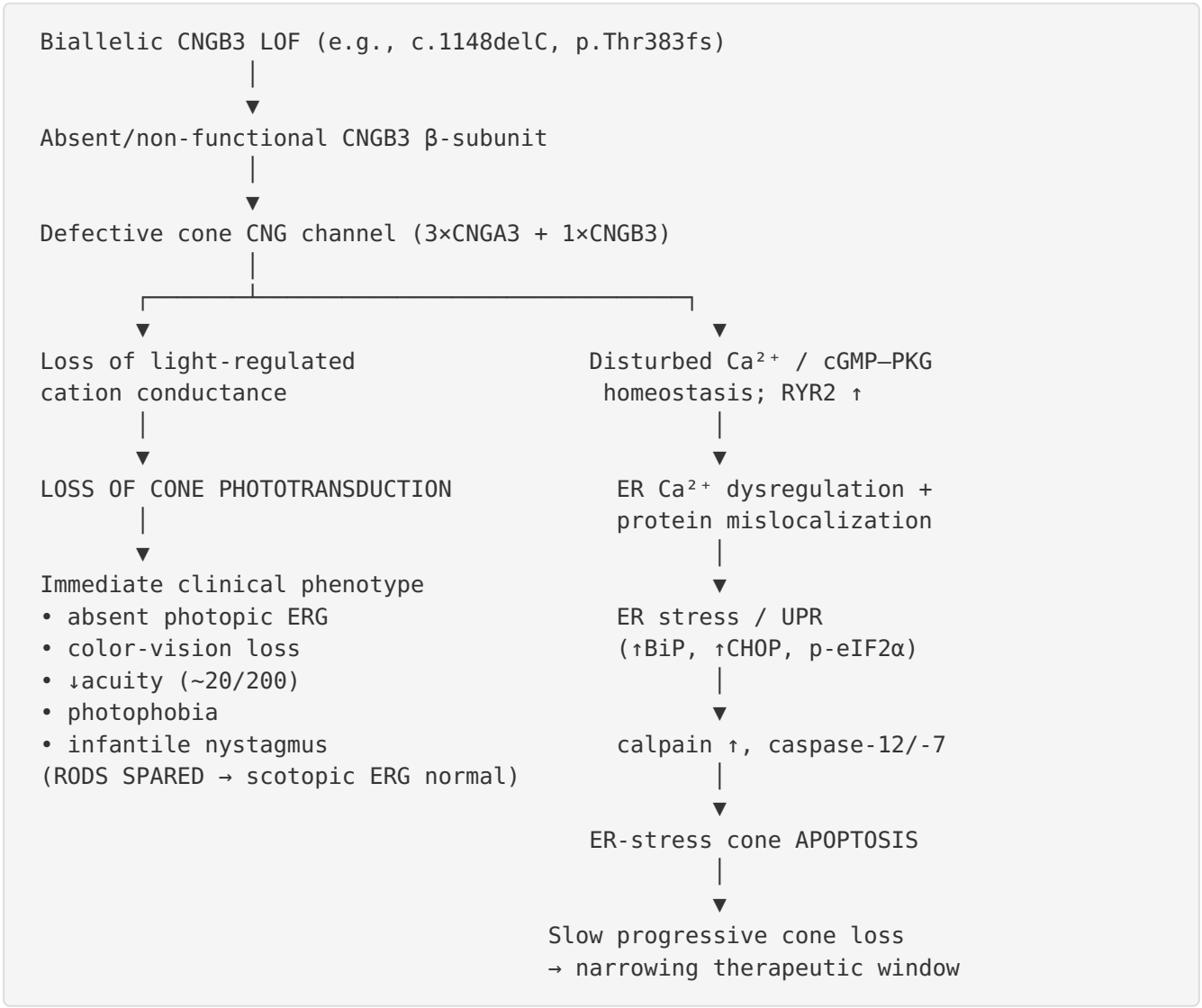
- **Dog (*Canis lupus familiaris*, NCBI Taxon 9615):** Naturally occurring **CNGB3 achromatopsia / "canine day blindness"** (e.g., Alaskan Malamute, German Shorthaired Pointer) is a validated large-animal model; AAV-CNGB3 rescues cone function (PMID: 29020838). Orthologous gene: canine *CNGB3*.
- **Mouse (*Mus musculus*, NCBI Taxon 10090):** *Cngb3* knockout recapitulates the disease (PMID: 21273547); ortholog *Cngb3* (MGI).
- **Comparative biology:** The cone CNG channel and phototransduction cascade are evolutionarily conserved across vertebrates, so disease mechanisms translate well between dog, mouse, and human.
- **Transmission / zoonosis:** Not applicable (genetic, non-transmissible).

15. Model Organisms

Model	Type	Genetic design	Phenotype recapitulation	Key reference
Cngb3$^{-/-}$ mouse	Mammalian, in vivo	Knockout	Early-onset, slowly progressive cone dysfunction; cone ERG ~50% WT by P15 → ~20% by 12 mo; cone density ~80%→~50%; opsin mislocalization; **rods spared**	PMID: 21273547
Cng$^{-/-}$;*Nrl*$^{-/-}$ cone-dominant mouse	Mammalian, in vivo	Compound KO (cone-enriched retina)	ER-stress/UPR-associated cone apoptosis (↑Grp78/BiP, ↑CHOP, caspase-12/-7 cleavage)	PMID: 22493484
CNGB3-mutant dog (day blindness)	Mammalian, large animal, natural	Naturally occurring CNGB3 mutation	Cone dysfunction; AAV-CNGB3 rescues cone function; proof-of-concept for human gene therapy	PMID: 29020838
Heterologous channel expression (e.g., Xenopus oocytes / HEK)	In vitro	Mutant CNG channel expression	Mechanistic dissection of channel gating/activity (e.g., spontaneous activity of CNGA3 R410W)	PMID: 35233102

Mouse strengths: faithful cone-functional phenotype and rod sparing; tractable for gene therapy and neuroprotection studies. **Mouse limitations:** rod-dominant retina (no macula/fovea), so the human foveal-cone and nystagmus phenotypes are not reproduced; the cone-dominant *Nrl*$^{-/-}$ cross partly addresses cone density. **Dog strengths:** cone-rich *area centralis*, eye size comparable to human, enabling subretinal AAV delivery testing and dose/toxicity assessment. **Resources:** MGI (mouse *Cngb3*); canine models via academic colonies; Alliance of Genome Resources for orthology.

Mechanistic Model / Interpretation



Upstream vs downstream. Upstream: CNGB3 LOF → non-functional channel (defines the disease and the immediate, largely non-progressive functional deficit). Downstream: Ca^{2+}/cGMP dysregulation → ER stress → apoptosis (drives the slow degenerative, progressive component). Therapeutically, **gene replacement targets the upstream lesion**, while **ER-stress/UPR modulation (e.g., SYVN1) targets the downstream degeneration** to widen the treatment window.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
28795510	CNGB3 mutation spectrum incl. CNVs, 552 patients	Human cohort	CNGB3 = major ACHM gene; 45.2% of 1074 families; β-subunit
15657609	CNGB3 accounts for 50% of AR achromatopsia	Human cohort	~50% of ACHM; c.1148delC >70% of alleles
17265047	p.T383fsX founder effect	Human genetics	Founder allele, 75% of disease alleles, shared haplotype
37172884	First-in-human AAV8-hCNGB3 (NCT03001310)	Human trial	Safe; no consistent efficacy at wk 24
35233102	ACHM channel mutation structure/function	In vitro	Aberrant channel activity → cone death
41867372	ACHM genotype/phenotype cohort	Human cohort	Phenotype frequencies; ERG pattern
39273686	Gene Therapy for Achromatopsia	Review	Prevalence 1/30,000; AR
28145975	Longitudinal foveal cone structure CNGB3	Human imaging	Structural stability over months
33562422	Longitudinal ACHM study	Human cohort	Slow functional progression over years
39365261	Foveal cone mosaic CNGA3	Human imaging	~2%/yr cone decline, like CNGB3
29020838	AAV5-CNGB3 in CNGB3-mutant dogs	Animal model	Functional rescue; dose-dependent toxicity
21273547	Cngb3$^{-/-}$ mouse	Animal model	Early-onset slow cone decline; rods spared
22493484	ER stress in CNG-channel deficiency	Animal/in vitro	ER-stress-associated cone apoptosis; >70% ACHM from CNGA3/CNGB3
32173907	RYR2 in cGMP/PKG cone degeneration	Animal/in vitro	Ca^{2+}/cGMP–PKG downstream mechanism

PMID	Title (abbrev.)	Evidence type	Supports
31182474	RYR2 & protein mislocalization	Animal/in vitro	ER Ca^{2+} dysregulation
39215566	SYVN1 overexpression preserves cones	Animal/in vitro	ER-stress modulation as neuroprotection

Limitations and Knowledge Gaps

1. **No consistent human efficacy for gene therapy.** The first-in-human AAV8-hCNGB3 trial showed safety but no consistent efficacy at 6 months (PMID: 37172884). Whether longer follow-up, different vectors/promoters/doses, younger patients, or cortical plasticity limits explain the result remains unresolved.
2. **Progression rate uncertainty.** Different modalities (OCT grade vs. BCVA vs. AOSLO cone density) give different pictures; precise CNGB3-specific decline rates and the optimal intervention window need larger natural-history cohorts.
3. **Downstream mechanism quantification in humans.** The Ca^{2+}/cGMP–PKG → ER-stress → apoptosis cascade is established mainly in mouse/in-vitro cone-dominant models; its quantitative contribution in human CNGB3 retina is inferred.
4. **No human omics signatures.** Transcriptomic/proteomic/metabolomic disease signatures for human CNGB3 retina were not identified.
5. **Model limitations.** Mice lack a fovea (the key human phenotype locus); the human nystagmus and foveal-hypoplasia phenotypes are not fully recapitulated.
6. **Population genetics detail.** Exact gnomAD allele/carrier frequencies for c.1148delC were not re-derived in this investigation.

Proposed Follow-up Experiments / Actions

1. **Refine the therapeutic window:** Prospective multi-year natural-history study combining AOSLO cone density, OCT EZ grading, BCVA, and chromatic psychophysics specifically in genotyped CNGB3 patients to define intervention cutoffs.

2. **Next-generation gene therapy:** Test improved cone-specific promoters, optimized AAV capsids/doses, and intervention in young children; incorporate perioperative immunosuppression informed by the canine chorioretinitis finding (PMID: 29020838).
3. **Combine replacement with neuroprotection:** Preclinically test AAV-CNGB3 plus ER-stress modulation (SYVN1/UPR-targeting) to preserve cones and widen the window (PMID: 39215566).
4. **Address cortical plasticity:** Pair gene therapy with structured visual training/rehabilitation to help a lifelong-deprived cone pathway "learn" to use restored signals.
5. **Human molecular profiling:** Apply single-cell/snRNA-seq and proteomics to patient-derived retinal organoids/iPSC-cones to map the human ER-stress signature and nominate adjunct drug targets.
6. **Carrier-screening and founder-allele programs:** Deploy targeted c.1148delC screening in high-frequency populations to support counseling and reproductive options.

Evidence source legend: Human clinical (cohort/trial/imaging), model organism (mouse/dog), in vitro (heterologous channel expression, retinal explant/organoid), and computational/review sources are distinguished in the Evidence Base table.
